IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 8 January 2025, accepted 20 January 2025, date of publication 23 January 2025, date of current version 10 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3532895

RESEARCH ARTICLE

A Comprehensive Approach to Intrusion Detection in IoT Environments Using Hybrid Feature Selection and Multi-Stage Classification Techniques

G. LOGESWARI, J. DEEPIKA ROSELIND, K. TAMILARASI, AND V. NIVETHITHA
School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu 600017, India
Corresponding author: G. Logeswari (logeswari.g@vit.ac.in)

ABSTRACT The rapid expansion of Internet of Things (IoT) devices has led to an increasingly complex threat landscape, challenging traditional Intrusion Detection Systems (IDS) to effectively handle the vast and diverse data generated by IoT networks. This paper presents a novel IDS that integrates Quantum-Inspired Particle Swarm Optimization (QIPSO) with Adaptive Neuro-Fuzzy Inference System (ANFIS) for optimized feature selection, followed by a multi-stage classification pipeline using Capsule Networks (CapsNets) and Attention Augmented-Recurrent Neural Networks (RNNs). The proposed approach addresses the limitations of existing methods by capturing both hierarchical and temporal dependencies in network traffic, improving the detection of subtle and advanced attacks. Evaluation on the TON-IoT and BOT-IoT datasets demonstrates that the proposed method significantly outperforms state-of-the-art techniques. Specifically, it achieves 98.83% accuracy, 98.56% precision, and 98.65% F-Measure on the TON-IoT dataset, and 98.6% accuracy, 98.5% precision, and 98.94% F-Measure on the BOT-IoT dataset, showcasing its superior performance over existing IDS models. This paper's key contribution lies in the integration of feature selection and classification techniques tailored for IoT environments, filling a critical gap in the state of the art and offering a more adaptive and efficient solution for real-time intrusion detection.

INDEX TERMS Enhanced recurrent neural network, adaptive neuro-fuzzy inference system, capsule networks, Internet of Things, intrusion detection system, quantum-inspired particle swarm optimization.

I. INTRODUCTION

The IoT encompasses an extensive network of interconnected devices that exchange and transmit data over the internet. These devices include everyday household gadgets like smart refrigerators and thermostats, as well as more advanced industrial sensors and wearable health tracking devices [1]. The rapid growth of IoT is largely driven by advancements in technology, including the proliferation of affordable sensors, improvements in wireless communication, and the widespread adoption of cloud computing. This expansion has significantly transformed both personal and industrial landscapes by enabling automation and fostering data-driven decision-making [2], [3], [4]. As a result, IoT has emerged as a crucial element of contemporary technology, embedding itself into numerous facets of everyday living and industrial operations [5]. The benefits of IoT are profound and varied. For individual users, IoT enhances convenience through smart home systems that automate tasks such as controlling lighting, adjusting heating, and managing security [6]. In the realm of smart cities, IoT contributes to more efficient traffic management, energy use optimization, and enhanced public safety. Industrial applications also see significant improvements with IoT, as real-time monitoring and optimization of processes lead to greater operational efficiency and reduced downtime [7], [8], [9], [10]. Moreover, the capacity to collect

The associate editor coordinating the review of this manuscript and approving it for publication was Sedat Akleylek.

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and examine vast amounts of data offers critical insights that foster innovation and enhance the quality of life in various fields.

Although IoT devices offer numerous benefits, their widespread adoption also presents considerable security risks. The incorporation of these devices into everyday activities and essential infrastructure increases the potential for cyber threats. Many IoT devices handle sensitive information, including financial data and personal health records, which makes them appealing targets for cybercriminals. Due to the interconnected structure of IoT networks, a security breach in a single device can jeopardize the entire system [11]. Therefore, implementing strong security protocols to safeguard data integrity, confidentiality, and availability is crucial for maintaining trust and protecting against potential risks.

The security landscape for IoT environments is fraught with potential threats and vulnerabilities. Common risks include data interception, where sensitive information is captured by unauthorized parties; device tampering, which can alter device functionality or compromise its integrity; and denial-of-service (DoS) attacks, which can disrupt the normal operation of devices and networks [12], [13], [14]. These vulnerabilities are exacerbated by the typically inadequate security measures found in numerous IoT devices, underscoring the necessity for robust security strategies to tackle these varied threats. The repercussions of security breaches in IoT environments can be significant and widespread. Unauthorized access to sensitive information may result in privacy infringements, financial setbacks, and reputational harm. In sectors like utilities and transportation, breaches can disrupt essential services and pose safety risks. Therefore, it is essential to uphold the integrity, confidentiality, and availability of data to preserve operational continuity and foster public confidence in IoT systems [15].

IDS face significant challenges in IoT environments due to the diverse nature of IoT devices, resource constraints, scalability issues, and dynamic network topologies. IoT networks consist of diverse devices, each with unique hardware capabilities, communication standards, and operating systems [16]. This variation makes it challenging to implement a single IDS that can efficiently oversee and secure all device types. Additionally, numerous IoT devices are constrained by limited processing capabilities, memory, and storage, making it challenging to implement traditional, resource-intensive security solutions. As the number of devices and the amount of data increase, conventional intrusion detection systems may face scalability issues, resulting in potential performance slowdowns. Moreover, IoT networks are extremely dynamic, with devices often entering and exiting the network, which complicates efforts to maintain effective and current security monitoring.

Traditional IDS primarily utilize two key methods: signature-based detection and anomaly-based detection. Signature-based IDS detects known threats by comparing network activity or system behavior against a database of predefined attack signatures [17]. This approach is effective at identifying attacks that match known patterns but is limited in its ability to detect new or unknown threats. Keeping the signature database updated with the latest attack signatures can be challenging, particularly in diverse and evolving IoT environments [18]. Anomaly-based IDS operate by establishing a baseline of typical behavior and identifying any deviations from this baseline as possible threats. While this method is effective for detecting new or unknown attacks that do not have established signatures, it can lead to a significant number of false positives due to the natural variability in normal activities. This problem is exacerbated in IoT environments, where the behavior of devices can vary greatly, complicating the management of false positives.

The shortcomings of conventional IDS methods in IoT environments emphasize the demand for more sophisticated techniques. Feature selection plays a vital role in enhancing IDS effectiveness by concentrating on the most significant features, which improves detection accuracy while minimizing false positives and false negatives [19], [20]. Additionally, this targeted approach helps optimize resource consumption, as fewer features lead to reduced computational and storage needs, making IDS deployment on resource-limited IoT devices more practical. Moreover, effective feature selection simplifies models, making them easier to interpret and enabling more informed decision-making.

Multi-stage classification further refines detection accuracy by processing data through several levels of analysis. This approach helps in reducing false positives by filtering out less likely threats and focusing on more probable ones, leading to more accurate detection. Additionally, multi-stage classification allows for a more detailed and precise analysis by capturing complex patterns and interactions between features. This capability improves the detection of sophisticated and novel attacks that might not be identified by simpler methods. The iterative nature of multi-stage classification also enhances the robustness of IDS against varying and noisy data, making the system more resilient to different types of attacks and operational changes.

Integrating hybrid approaches, such as advanced feature selection techniques and multi-stage classification pipelines, provides a comprehensive solution for improving IDS performance. Hybrid feature selection techniques, like QIPSO, enable the identification of optimal feature subsets, enhancing the efficiency and effectiveness of IDS. Implementing multi-stage classification pipelines with advanced models, such as CapsNets and Attention-Augmented RNNs, allows for more accurate and adaptive threat detection. This integration addresses the limitations of traditional methods, resulting in a more robust and effective security system capable of securing diverse and dynamic IoT environments.

A. MOTIVATION

In the fast-paced domain of network security, the ability to detect and mitigate threats both efficiently and accurately is paramount. Traditional IDS encounter difficulties in

handling the vast amounts of network traffic, high-dimensional datasets, and the growing complexity of cyber-attacks. The proliferation of IoT devices has further complicated network traffic, necessitating more advanced and adaptive intrusion detection solutions. The proposed system tackles these challenges by incorporating cutting-edge techniques in preprocessing, feature selection, and multi-stage classification. It leverages advanced algorithms such as QIPSO and CapsNets, alongside powerful models like Attention-Augmented RNNs and ANFIS, to improve the precision and efficiency of intrusion detection and classification.

B. OBJECTIVES

The objectives of the proposed system are to:

- Develop a solution capable of efficiently processing large volumes and high-speed network traffic, ensuring that the IDS can scale with growing network demands.
- Implement SMOTE to generate synthetic samples for minority classes, improving the model's ability to detect rare but significant anomalies and ensuring balanced detection performance.
- Integrate a hybrid approach using QIPSO and ANFIS to identify and refine the most relevant features, addressing the challenges of high-dimensional data and feature interaction complexities.
- Utilize a multi-stage classification pipeline combining CapsNets and Attention-Augmented RNNs to effectively handle both hierarchical patterns and temporal dependencies, enhancing adaptability to new and sophisticated attack vectors.

C. CONTRIBUTIONS

This research addresses the critical challenges of intrusion detection in IoT environments by introducing innovative methodologies that enhance feature selection and classification accuracy. The proposed system integrates advanced optimization and deep learning techniques to tackle the complexity of high-dimensional data and subtle attack patterns effectively.

- Introduced a two-tier method combining QIPSO for efficient feature exploration and ANFIS for refined selection, representing a novel approach in IoT IDS.
- Developed a unique pipeline using CapsNets for hierarchical attack detection and Attention Augmented-RNN for detailed attack classification, integrating spatial and temporal learning.
- Combined CapsNets for spatial analysis and attention-based RNNs for temporal dependencies, uniquely optimizing detection of complex IoT attacks.

D. PAPER ORGANIZATION

The paper is structured as follows: Section I, Introduction, sets the stage by outlining the IoT environment, highlighting security challenges, identifying research gaps, and stating the objective to develop an advanced IDS system using novel techniques. Section II, Related Work, reviews existing IDS methods and advancements in feature selection and multi-stage classification. Section III, Theoretical Framework, explains key concepts such as Quantum-Inspired Particle Swarm Optimization, Adaptive Neuro-Fuzzy Inference System, Capsule Networks, and Attention-Augmented Recurrent Neural Networks. Section IV, Proposed System, details the architecture and components of the new IDS system incorporating these advanced techniques. Section V, Performance Evaluation, describes the metrics and methods for evaluating the proposed system, presents results, and compares its performance with traditional methods. Finally, Section VI, Conclusion and Future Work, summarizes the findings, discusses the system's impact, and suggests directions for future research.

II. RELATED WORK

Intrusion detection in IoT environments poses considerable challenges due to the wide variety of devices, limited computational resources, and the requirement for immediate detection. To tackle these issues, researchers have investigated various feature selection and machine learning techniques. This literature review examines recent progress in hybrid feature selection techniques and multi-stage classification approaches for intrusion detection within IoT networks. Multi-stage classification techniques improve detection performance by utilizing sequential classifiers. A prevalent approach is to utilize a two-stage hybrid IDS: the initial stage focuses on binary classification to distinguish between normal and attack traffic, while the subsequent stage utilizes advanced detection techniques to investigate the identified attacks. Additionally, hybrid approaches integrate both filter and wrapper feature selection techniques, enhancing feature relevance through multiple stages. Another strategy involves using anomaly detection in the first stage, followed by misuse detection in the second. These techniques collectively improve detection accuracy and reduce computational costs compared to single-stage approaches, addressing the unique challenges of IoT security.

Nazir et al. [19] introduced a hybrid feature selection approach for IDS in IoT environments. It combines statistical filter methods, like Chi-Square and Pearson's Correlation, with meta-heuristic algorithms to enhance feature relevance while reducing computational complexity. Evaluated on various IoT datasets, the framework outperforms traditional methods in detection accuracy and feature reduction. The authors highlight the importance of choosing suitable feature selection techniques based on specific IoT application needs, concluding that this hybrid mechanism significantly improves cyber-attack detection in IoT networks. Sharma et al. [20] proposed a two-stage hybrid intrusion detection system that enhances cyber-attack detection. In the first stage, the system employs a binary classification to differentiate between normal and attack traffic. The second stage further analyzes traffic records flagged as attacks, applying advanced detection techniques to improve accuracy. The

hybrid approach integrates various machine learning methods to optimize performance and reduce false positives. The authors demonstrate that this two-stage methodology effectively enhances the detection capabilities of IDSs, making it suitable for dynamic network environments.

Sanju [21] proposed a hybrid approach using filter and wrapper methods for feature selection combines the strengths of both techniques to enhance classification performance. Initially, a filter method quickly selects relevant features based on statistical measures, reducing the feature set size. Subsequently, a wrapper method refines this selection by evaluating the performance of machine learning models on the chosen features. This dual-stage process improves classification accuracy while maintaining computational efficiency, making it effective for various applications, including sentiment analysis and cyber-attack detection in IoT environments. Nivaashini et al. [22] explored feature selection methods such as Information Gain, Chi-Squared statistics, and Correlation-based Feature Selection to enhance intrusion detection in wireless IoT networks. It proposes a hybrid IDS framework combining anomaly detection through K-means clustering and misuse detection using a Random Forest classifier. The approach aims to reduce false positive rates while improving detection accuracy, demonstrating high performance on the Aegean Wi-Fi Dataset across various attack classes, including impersonation and flooding.

Walling and Lodh [23] explored the widespread adoption of IoT devices has brought new security challenges, intensifying the focus on IDS among researchers. A novel feature selection methodology for anomaly-based NIDS is proposed, employing 1-way ANOVA and Pearson correlation as initial filters to identify relevant features. The optimal features are then extracted using set theory principles, ensuring only the most informative attributes are retained for subsequent analysis, addressing the challenges posed by IoT IDS systems due to their functional and physical diversity. Hajisalem and Babaie [24] proposed a novel IDS that integrates the Artificial Bee Colony (ABC) algorithm with an Adaptive Feature Selection (AFS) method. This hybrid system effectively identifies both misuse and anomaly attacks in network traffic. By optimizing feature selection, the approach enhances detection accuracy while minimizing false positives. The proposed system shows substantial enhancements in performance, making it well-suited for the dynamic and varied landscapes of IoT environments. It effectively meets the rising demand for strong security solutions amid escalating cyber threats.

Khan et al. [25] introduced an innovative deep learning model for flood detection on edge devices. This model employs a hybrid strategy that integrates a convolutional neural network (CNN) with a deep ResNet classifier. To achieve image segmentation and feature extraction, the authors introduce a cubic chaotic map weighted k-means clustering method, in conjunction with feature extraction based on vegetation indices. The model's performance is further improved by employing a CHHSSO-based training approach, which optimizes the weights to attain high levels of accuracy. Experiments on satellite imagery demonstrate the model's effectiveness in detecting floods while maintaining computational efficiency suitable for edge deployment. Shyaa et al. [26] introduced an advanced IDS that utilizes data stream classification and addresses concept drift through incremental learning. By integrating genetic programming, the system adapts to evolving attack patterns in real-time, enhancing detection accuracy. The proposed method demonstrates improved performance in identifying intrusions while efficiently managing the dynamic nature of data streams, making it suitable for modern cybersecurity challenges.

Wu et al. [27] proposed an innovative intrusion detection model utilizing CNNs to automatically select traffic features from raw datasets. It addresses the challenge of imbalanced datasets by adjusting the cost function weight coefficients based on class distribution. The model demonstrates high detection accuracy and efficiency, making it suitable for large-scale networks. By eliminating the need for manual feature engineering, this end-to-end approach enhances the effectiveness of IDSs in identifying various attack patterns within massive network environments. Ayad et al. [28] present a hybrid approach for efficient feature selection in anomaly-based intrusion detection for IoT networks. It employs a two-stage process. The first stage uses statistical filter methods like Chi-Square and Pearson's Correlation to rank features based on their relevance. This quickly selects an initial subset of important features. The second stage applies a wrapper method using meta-heuristic algorithms to further optimize the feature subset selected in the first stage. This ensures the final set of features provides the best detection performance. By integrating the advantages of both filter and wrapper methods, the hybrid approach attains high accuracy while preserving computational efficiency, rendering it ideal for real-time intrusion detection in resource-limited IoT environments.

Guerra-Manzanares et al. [29] explored hybrid feature selection models for detecting botnets in IoT networks, emphasizing improvements in detection accuracy and efficiency. Their approach combines various feature selection techniques, including filter and wrapper methods, to pinpoint the most pertinent features from network traffic data. By refining the feature selection process, the model seeks to enhance the performance of machine learning algorithms in identifying botnet attacks while considering the limitations of IoT environments, such as restricted bandwidth and computational resources. The approach demonstrates significant improvements in detection rates, making it a valuable contribution to IoT security. Liu and Shi [30] introduce a hybrid IDS that combines a Genetic Algorithm (GA) for feature selection with a Random Forest classifier. The GA optimizes the identification of relevant features from network traffic data, thereby improving detection accuracy. The Random Forest model then classifies the selected features to identify

potential intrusions. This approach effectively reduces false positives and improves overall system performance, making it suitable for dynamic network environments.

Maseno and Wang [31] presented a hybrid wrapper feature selection approach that combines a GA with an extreme learning machine (ELM) classifier for intrusion detection. The GA optimizes the selection of relevant features from network traffic data, while the ELM model assesses the fitness of these selected feature subsets. This combined approach aims to improve detection accuracy and reduce false positives by identifying the most informative features. The method shows superior performance compared to using GA or ELM individually, making it well-suited for real-time intrusion detection in IoT environments. In a different study, Okpomo Okpu et al. [32] propose an innovative approach that integrates various machine learning techniques for effective intrusion detection and mitigation in smart healthcare IoT environments. Their model combines several algorithms to enhance detection accuracy and minimize false positives. By leveraging real-time data analysis, it addresses the unique security challenges posed by IoT devices in healthcare settings, ensuring robust protection against cyber threats while maintaining system performance and reliability. The proposed solution demonstrates significant improvements in identifying and mitigating potential intrusions.

Gurumurthy et al. [33] proposed a novel feature selection method that combines Pigeon Inspired Optimization (PIO) and Gray Wolf Optimization (GWO) to enhance intrusion detection systems. The hybrid approach aims to optimize feature selection by leveraging the strengths of both algorithms, improving detection accuracy and reducing false positives. Evaluated on standard datasets, the method demonstrates significant improvements in identifying network attacks, making it effective for real-time intrusion detection in IoT environments. Prasath et al. [34] proposed an optimal secure defense mechanism to combat DDoS attacks in IoT networks through feature optimization and IDSs. Their hybrid approach integrates both filter and wrapper feature selection methods to identify the most pertinent network traffic features. These selected features are subsequently utilized to train machine learning models for precise detection of DDoS attacks. The proposed mechanism seeks to improve detection accuracy while reducing false positives, making it effective for protecting IoT networks against evolving DDoS threats.

Sajid et al. [35] introduced a hybrid intrusion detection model that integrates machine learning and deep learning techniques to improve detection accuracy. The model utilized XGBoost and CNN for feature extraction, followed by Long Short-Term Memory (LSTM) networks for classification. This hybrid approach aimed to address the challenges posed by increasing data volume, feature dimensions, and new attack types in cloud-based systems. Experiments on benchmark datasets demonstrated high detection rates and low false acceptance rates, proving the effectiveness of the proposed model.

The literature on hybrid feature selection and multi-stage classification techniques for intrusion detection in IoT environments reveals significant advancements in enhancing detection accuracy and efficiency. Various studies demonstrate the effectiveness of combining statistical methods with machine learning algorithms, such as Genetic Algorithms, Random Forests, and Convolutional Neural Networks. These approaches address the challenges posed by diverse IoT applications and evolving attack patterns, ultimately improving system performance while minimizing false positives. The integration of these techniques not only optimizes feature selection but also ensures robust security solutions tailored for dynamic and resource-constrained IoT networks, highlighting the critical need for ongoing research in this area.

Table 1 highlights the methodologies, feature selection techniques, classification approaches, and key contributions of various studies addressing intrusion detection challenges in IoT networks.

Despite significant advancements in IDS, several research gaps continue to challenge their effectiveness. Traditional IDS solutions often face difficulties in scalability, struggling to manage the vast volumes and high-speed data traffic typical of modern networks. Furthermore, high-dimensional feature spaces pose a challenge, as conventional feature selection methods frequently fall short in capturing intricate interactions and hierarchical patterns among features. Class imbalance remains a critical issue, with many systems inadequately addressing the disproportion between normal and attack traffic, which undermines the detection of rare but impactful attacks. Additionally, existing IDS models often rely on static, rule-based approaches that lack adaptability to evolving and sophisticated threats. The integration of cutting-edge algorithms such as QIPSO and CapsNets is often limited, missing opportunities for enhanced feature selection and classification. Real-time processing capabilities can also be insufficient, resulting in delays that hinder timely threat mitigation. Moreover, the complexity of advanced machine learning models used in IDS can lead to challenges in explainability and interpretability, complicating the understanding and trust of the system's decisions. Resource-intensive implementations further restrict their use in environments with limited computational capacity. The proposed system addresses these gaps by incorporating scalable preprocessing methods, advanced hybrid feature selection using QIPSO and ANFIS, and a multi-stage classification pipeline with CapsNets and Attention-Augmented RNNs. This approach enhances scalability, adaptability, and real-time processing, while improving feature selection and classification accuracy, ultimately advancing IDS performance and effectiveness.

III. THEORETICAL FRAMEWORK

In the domain of IDS for IoT environments, the challenge of accurately detecting and mitigating security threats is compounded by the complexity and dynamism of the systems involved. To address these challenges effectively, advanced computational and machine learning techniques are essential.

TABLE 1. Summary of related works on intrusion detection in IoT environments.

Study	Feature Selection Techniques	Classification Approach	Key Contributions
Nazir et al. [19]	Chi-Square, Pearson's Correlation, Meta-heuristic algorithms	Hybrid ML-based IDS	Hybrid approach improves detection accuracy and reduces computational complexity.
Sharma et al. [20]	N/A	Two-stage hybrid IDS: Binary classification + Advanced detection	Enhanced detection with reduced false positives; effective for dynamic IoT environments.
Sanju [21]	Filter and Wrapper Methods	Machine Learning	Dual-stage selection improves classification performance while maintaining efficiency.
Nivaashini et al. [22]	Information Gain, Chi-Square, Correlation-based Selection	K-Means (Anomaly Detection) + Random Forest	Reduced false positives; high accuracy in detecting impersonation and flooding attacks.
Walling et al. [23]	1-Way ANOVA, Pearson Correlation	Anomaly-Based NIDS	Feature selection optimized using set theory principles; suitable for diverse IoT environments.
Hajisalem et al. [24]	Artificial Bee Colony (ABC) + Adaptive Feature Selection (AFS)	Hybrid IDS (Misuse + Anomaly)	High detection accuracy; low false positives; adaptable for dynamic IoT environments.
Liu et al. [30]	Genetic Algorithm (GA)	Random Forest	Optimized feature selection with reduced false positives and improved detection accuracy.
Maseno et al. [31]	Genetic Algorithm (GA) + Extreme Learning Machine (ELM)	Hybrid IDS	Improved accuracy; suitable for real-time intrusion detection in resource-constrained environments.
Sajid et al. [35]	XGBoost for feature extraction	CNN + LSTM	Handles large data volumes and evolving attack types with high detection rates.
Prasath et al. [34]	Filter and Wrapper Methods	Machine Learning Models	Effective DDoS detection with high accuracy and reduced false positives.
Gurumurthy et al. [33]	Pigeon Inspired Optimization (PIO) + Grey Wolf Optimization (GWO)	Hybrid IDS	Enhanced accuracy in real-time attack detection.

This section explores four key theoretical frameworks that form the foundation of modern IDS approaches: QIPSO, ANFIS, CapsNets, and Attention-Augmented RNNs. Each of these frameworks offers distinct advantages in optimizing feature selection, enhancing classification accuracy, and improving the overall robustness of IDS.

A. QUANTUM-INSPIRED PARTICLE SWARM OPTIMIZATION

QIPSO is an advanced optimization technique that extends the traditional PSO algorithm by integrating concepts inspired by quantum computing [36]. In PSO, a group of particles, each representing a potential solution, navigates the solution space by updating their positions and velocities. These adjustments are influenced by both the particle's individual experience and the experience of its neighbors. The goal is to find the optimal solution by balancing exploration and exploitation. QIPSO enhances this process by incorporating quantum-inspired mechanisms such as quantum bits (qubits) and quantum gates. These mechanisms allow for more sophisticated manipulation of particle states, improving the algorithm's ability to explore diverse regions of the solution space [37]. For example, quantum-inspired techniques enable particles to exhibit quantum superposition and entanglement, which can help them escape local optima and explore the global search space more effectively. This is particularly beneficial in high-dimensional feature selection tasks where, identifying the most relevant features among many is crucial. In IDS, QIPSO can be employed to optimize feature selection processes, thereby improving the accuracy and efficiency of intrusion detection.

B. ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM

The ANFIS combines neural networks and fuzzy logic to create a hybrid system capable of modelling complex, uncertain, or imprecise data. ANFIS uses neural networks to adaptively

tune the parameters of a fuzzy inference system, which applies fuzzy logic principles to represent and process data. This combination allows ANFIS to handle various types of data and uncertainties more effectively than either approach alone [38]. In an ANFIS model, fuzzy logic provides a way to represent knowledge and reasoning with linguistic variables, which can handle vague or imprecise information. The neural network component is used to train the fuzzy system, adjusting the membership functions and rule parameters based on input data. This adaptation improves the system's ability to model and predict complex relationships [39]. In the context of IDS, ANFIS can be employed to classify network traffic or system behaviors by learning from historical data and adapting to new patterns, enhancing the system's capability to detect anomalies and potential intrusions accurately.

C. CAPSULE NETWORKS

CapsNets represent a novel approach to neural network design aimed at overcoming some of the limitations of traditional CNNs. CapsNets use capsules—groups of neurons that work together to represent specific features or patterns—instead of individual neurons. Each capsule outputs a vector that encodes the probability and the properties of the detected feature, such as its pose and orientation. The key innovation of CapsNets is the dynamic routing mechanism between capsules. This mechanism allows capsules to send their outputs to other capsules in a hierarchical manner, based on the agreement between the predicted and actual feature properties [40], [41]. This dynamic routing helps CapsNets capture complex patterns and spatial relationships more effectively than CNNs, which often struggle with issues like pose variation and part-whole relationships. In IDS, CapsNets can enhance the detection of sophisticated attacks by improving the network's ability to recognize intricate patterns and relationships in data.

D. ATTENTION-AUGMENTED RECURRENT NEURAL NETWORK

Attention-Augmented RNNs enhance traditional RNN architectures by integrating attention mechanisms. While RNNs are designed to process sequential data by maintaining a state that reflects information from previous time steps, making them effective for time-series analysis and sequence modeling, the addition of attention mechanisms allows these networks to focus on different parts of the input sequence more dynamically. This integration improves the model's ability to capture relevant information and handle long-range dependencies in sequential data. The attention mechanism enhances RNNs by enabling the model to concentrate on the most pertinent segments of the input sequence for the given task. Rather than treating the entire sequence with uniform importance, attention mechanisms allocate varying weights to different parts of the sequence, emphasizing the most informative segments [42]. This dynamic weighting process improves the model's capacity to capture long-range dependencies and contextual information more effectively. In the realm of IDS, Attention-Augmented RNNs can improve the detection of temporal patterns and anomalies by paying closer attention to critical segments of the data. This capability is particularly useful for identifying sequential attack patterns or changes in network behavior over time, leading to more accurate and responsive intrusion detection. Each of these theoretical frameworks offers valuable insights and capabilities that contribute to enhancing IDS for IoT environments. By integrating these advanced techniques, IDS can achieve improved accuracy, robustness, and efficiency in detecting and mitigating security threats.

IV. PROPOSED METHODOLOGY

The proposed IoT-based intrusion detection system combines advanced techniques in preprocessing, feature selection, and multi-stage classification to achieve high accuracy and robustness. Preprocessing ensures data quality by standardizing, normalizing, and reducing noise, while addressing class imbalances using SMOTE to generate synthetic samples for underrepresented classes. For feature selection, QIPSO efficiently explores the feature space to identify optimal subsets, while ANFIS fine-tunes these subsets using fuzzy logic and error minimization to enhance model performance.

The classification stage employs a two-step process. First, Capsule Networks (CapsNets) detect whether network traffic is normal or an attack, effectively capturing hierarchical patterns in the data. Second, an Attention-Augmented RNN is used to analyze temporal dependencies, enabling the precise identification of specific attack types. This multi-stage approach improves the system's ability to classify both known and unknown IoT traffic scenarios accurately.

Hyperparameter tuning through a grid search method further optimizes the system's performance. Parameters for QIPSO (e.g., swarm size and inertia weight), ANFIS (e.g., membership functions and learning rate), CapsNets (e.g., number and dimensions of capsules), and Attention-Augmented RNN (e.g., hidden layer size and attention mechanism) are adjusted to ensure the best configuration. This comprehensive methodology results in a highly adaptive and efficient system capable of handling complex IoT environments.

A. DATA PRE-PROCESSING

Data preprocessing is a critical phase that transforms raw sensor data (e.g., source and destination IPs, ports, protocol types) into a format suitable for effective analysis and anomaly detection. The preprocessing process begins with data standardization and normalization.

Standardization is applied to adjust data so that it has a mean of zero and a standard deviation of one, ensuring that features from different devices are comparable. This is achieved using (1).

$$x_{i,standardized} = \frac{x_i - \mu}{\sigma} \tag{1}$$

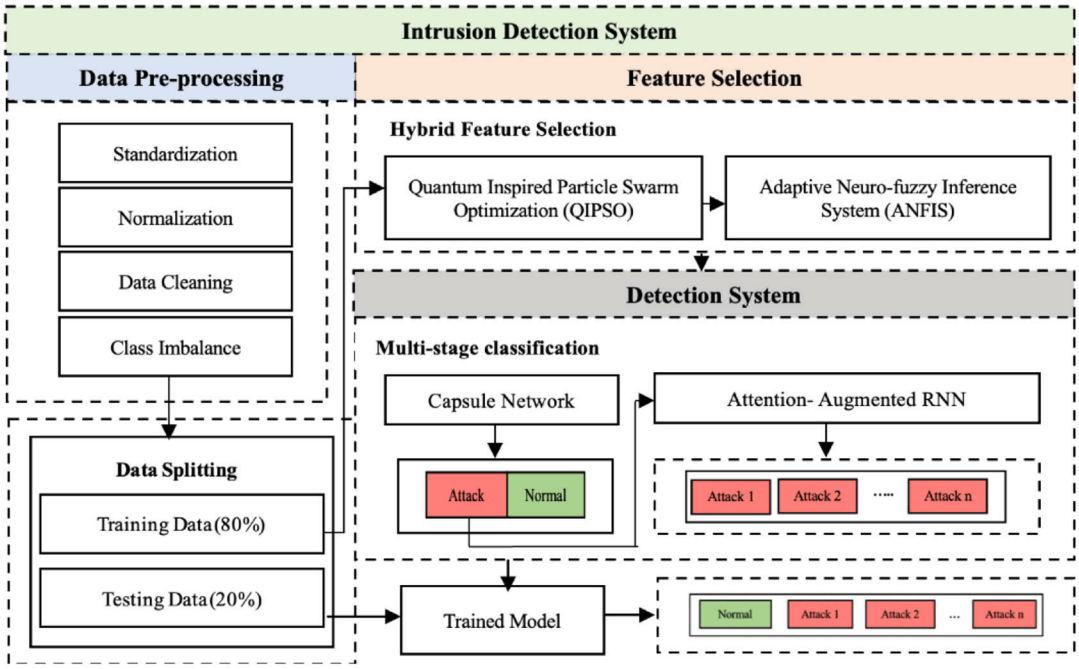

FIGURE 1. Architectural diagram of the proposed system.

where, x_i represents the raw value of the feature, μ denotes the mean of the feature values, and σ is the standard deviation of the feature values. Standardization helps to prevent features with larger ranges from overshadowing others in the learning process, ensuring that each feature contributes equally to the model.

Following standardization, normalization scales the data to a specific range, typically [0,1], to ensure that all features are on a similar scale. Normalization helps to mitigate any bias introduced by the original scale of the data. The normalization formula is given in (2).

$$x_{i,normalized} = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{2}$$

where, x_{min} *and* x_{max} are the minimum and maximum values of the feature, respectively.

The next step is data cleaning, which addresses issues such as missing values and noise. Missing values in IoT data are commonly handled using mean imputation, which replaces missing values with the mean of the feature. The imputation formula is given in (3).

$$x_{imputed} = mean(x) \tag{3}$$

where, *mean* (x) represents the average of the feature values. This method maintains the dataset's integrity and ensures that missing values do not skew the analysis.

To manage noise in the data, moving average smoothing is employed. This technique reduces short-term fluctuations and emphasizes longer-term trends by averaging data points within a window. The moving average is calculated as given in (4).

$$x_{smoothed} = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{4}$$

where, N is the number of points in the smoothing window and x_i refers to each data point within the chosen window that is being averaged to produce the smoothed value $x_{smoothed}$. This method effectively filters out random noise, providing a clearer and more reliable signal from the sensor data.

Finally, to address class imbalance, SMOTE is used. SMOTE generates synthetic samples for minority classes to balance the dataset. The synthetic sample generation is defined by (5).

$$x_{new} = x_{minority} + \lambda \times (x_{neighbor} - x_{minority}) \tag{5}$$

where, $x_{minority}$ is a sample from the minority class, $x_{neighbor}$ is a nearest neighbor of the minority class sample, and λ is a random number between 0 and 1. SMOTE helps to increase the representation of less frequent attack types, improving the model's ability to detect rare but critical anomalies.

B. DATA SPLITTING

The dataset is divided into two primary subsets, with 80% designated as the training set for model development and the remaining 20% reserved as the test set for performance evaluation. This approach ensures the model is trained effectively and evaluated on unseen data. To maintain the class distribution in both subsets, a stratified splitting technique is employed. This ensures that the class proportions in

real-world IoT scenarios are accurately represented, leading to a more reliable assessment of the model's performance.

C. HYBRID FEATURE SELECTION

The hybrid feature selection process integrates QIPSO and Adaptive Neuro-Fuzzy Inference System ANFIS to identify the most relevant features efficiently. In recent research, the integration of Quantum-Inspired Particle Swarm Optimization (QIPSO) and Adaptive Neuro-Fuzzy Inference System (ANFIS) has been explored to enhance feature selection processes in various domains, including IoT and network intrusion detection. The synergy between QIPSO's global optimization capabilities and ANFIS's adaptability for non-linear decision-making has demonstrated efficiency in identifying relevant features, which are essential for building accurate predictive models

1) QUANTUM-INSPIRED PARTICLE SWARM OPTIMIZATION

Initially, QIPSO is utilized to explore and select feature subsets. This algorithm starts by initializing a population of particles, each representing a candidate feature subset as a binary vector. Each bit in this vector indicates whether a particular feature is included (1) or excluded (0). The positions x_i and velocities v_i of these particles are updated iteratively according to quantum-inspired rules. The velocity update equation is given in (6).

$$v_i^{(t)} = \omega v_i^{(t-1)} + c_1 r_1 (p_i - x_i^{(t-1)}) + c_2 r_2 (g - x_i^{(t-1)}) \quad (6)$$

where, ω is the inertia weight controlling the influence of previous velocities, $v_i^{(t-1)}$ is velocity of particle i at the previous time step, c_1 and c_2 are cognitive and social coefficients, r_1 and r_2 are random values between 0 and 1, p_i represents the personal best position of particle i, g denotes the global best position across all particles and $x_i^{(t-1)}$ denotes the current position of particle i.

The position update of each particle is performed using (7).

$$x_i^{(t)} = x_i^{(t-1)} + v_i^{(t)} \quad (7)$$

where, $x_i^{(t-1)}$ is the position vector of particle i at the previous time step t−1. $v_i^{(t)}$ is the velocity of particle i at the current time step t.

To maintain binary values in the feature subset, a threshold function is applied as given in (8).

$$x_{ij}^{(t)} = \begin{cases} 1, & if\ x_{ij}^{(t)} \geq 0.5 \\ 0, & otherwise \end{cases} \quad (8)$$

The fitness of each feature subset is evaluated by training a classifier on the selected features and measuring its performance, such as classification accuracy as denoted in (9).

$$F\ (x_i) = Accuracy\left(Classifier\left(x_{ij}^{(t)}\right)\right) \quad (9)$$

Once QIPSO identifies promising feature subsets, these subsets are passed to ANFIS for further refinement.

2) ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM (ANFIS)

ANFIS employs fuzzy logic to model the relevance of features through a set of fuzzy rules and membership functions. The relevance of features is modelled by fuzzy rules as given in (10).

$$R_i{:}IF\ x_1\ is\ A_{1i}\ AND\ x_2\ is\ A_{2i}\ THEN\ y_i \quad (10)$$

where, A_{1i} and A_{2i} are fuzzy sets representing the membership functions of features, and y_i is the relevance score of the feature subset.

ANFIS is trained by minimizing the error between the actual relevance scores and those predicted by the fuzzy model. The error is calculated as denoted in (11).

$$E = \frac{1}{N}\sum_{j=1}^{N}(y_j - \hat{y}_j)^2 \quad (11)$$

where, E is the mean squared error, N is the number of samples, y_j is the actual relevance score of feature j and $\hat{y}_j$ is the predicted relevance score from ANFIS.

The ANFIS model adjusts the parameters of the fuzzy rules and membership functions to minimize this error, refining the relevance scores of features. The final feature subset is selected based on these refined scores, ensuring that the chosen features are both relevant and efficient for enhancing the IDS's detection capabilities. This hybrid approach leverages QIPSO's ability to explore and identify potential feature subsets and ANFIS's capability to refine and model feature relevance, providing a robust solution for feature selection in complex IoT environments.

D. MULTISTAGE CLASSIFICATION

In the first stage of the multi-stage classification pipeline, the system begins with CapsNets to perform initial traffic classification, distinguishing between normal and attack traffic. Attack traffic is then further analyzed by an Attention-Augmented RNN to classify the specific type of attack.

1) CAPSULE NETWORKS

CapsNets are employed to extract hierarchical features and classify network traffic into two categories: normal and attack. CapsNets utilize capsules, which are groups of neurons that capture various aspects of the input data through their output vectors. The capsules work by detecting complex patterns and relationships in the data, crucial for distinguishing between normal and attack traffic.

Feature Extraction: Each capsule computes an output vector u_{ij} from the input features, which represents the presence of a particular feature in the data. This is achieved using dynamic routing algorithms to ensure that only the most relevant features are forwarded to higher-level capsules.

Dynamic Routing: The coupling coefficients c_{ij} are updated iteratively to determine the influence of lower-level capsules on higher-level capsules. The updated output vector

Algorithm 1 Hybrid Feature Selection

Input: N ← Number of particles, D ← Dimensionality of feature space, ω ← Inertia weight, c1 ← Cognitive coefficient, c2 ← Social coefficient, T ← Number of iterations, Initial feature set (set of all features)
Output: Final optimized feature subset.
Begin:
for i ← 1 to N
  x_i ← Binary vector of length D (0s and 1s)
  v_i ← Velocity vector initialized to zeros
  Train classifier using features selected by x_i
  F(x_i) ← Fitness based on performance metrics
  if F(x_i) > F(p_i) then
    $p_i \leftarrow x_i$
  end if
  g ← Best position among all p_i
end for
for t ← 1 to T
  for i ← 1 to N
    $v_i^{(t)}$ ← Update velocity based on ω, c1, c2, p_i, and g
    $x_i^{(t)}$ ← Update position using $v_i^{(t)}$ and apply binary threshold
    F($x_i^{(t)}$) ← Re-evaluate fitness
    if F($x_i^{(t)}$) > F(p_i) then
      $p_i \leftarrow x_i^{(t)}$
    end if
    if F(p_i) > F(g) then
      g ← p_i
    end if
  end for
end for
for each x_i identified by QIPSO
  Define fuzzy rules and membership functions
  Train ANFIS using x_i
    Minimize error between actual and predicted relevance scores
  Refine relevance scores for features using trained ANFIS
  Select features with highest relevance scores
end for
end

v_j is given by (12).

$$v_j = squash(\sum_i c_{ij} u_{ij}) \tag{12}$$

where, the squash function denoted in (13) normalizes v_j.

$$squash\,(v) = \frac{||v||^2}{1 + ||v||^2} \cdot \frac{v}{||v||} \tag{13}$$

Classification Decision: The final output from the CapsNets is used to classify the traffic into normal or attack. This decision is made by applying a soft-max function to the capsule outputs to obtain probabilities as denoted in (14).

$$y = softmax\left(Wv_j + b\right) \tag{14}$$

where, W and b are the weights and biases of the classification layer, respectively, and the soft-max function is given in (15).

$$softmax\,(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{15}$$

2) ATTENTION-AUGMENTED RNN

The attack traffic identified by CapsNets is then passed to an Attention-Augmented RNN for further classification to determine the specific type of attack. The RNN processes the sequential nature of the attack data, and the attention mechanism allows it to focus on significant time steps.

Temporal Modeling: The RNN processes the sequence of attack data to capture temporal dependencies. Each hidden state h_t is updated using (16).

$$h_t = RNN(x_t, h_{t-1}) \tag{16}$$

where, x_t is the input feature at time step t, and h_{t-1} is the previous hidden state.

Attention Mechanism: The attention mechanism computes attention scores α_t to weigh the importance of different time steps using (17).

$$\alpha_t = \frac{\exp(score\,(h_t, s_{t-1}))}{\sum_k \exp\,(score(\,(h_k, s_{t-1}))} \tag{17}$$

where, score (h_t, s_{t-1}) is the relevance score between the current hidden state and the previous state.

Context Vector: The context vector c_t is computed as a weighted sum of all hidden states as shown in (18).

$$c_t = \sum_k \alpha_k \mathrm{h}_k \tag{18}$$

Final Classification: The context vector c_t is used to classify the type of attack through a final soft-max layer as given in (19).

$$y = softmax(W_{c_t} + b) \tag{19}$$

In this multi-stage pipeline, CapsNets provide a robust initial classification by focusing on complex feature relationships, while the Attention-Augmented RNN refines the classification by leveraging temporal patterns and focusing on key moments in the attack data. This approach ensures that normal traffic is efficiently filtered out, and specific attack types are accurately identified.

V. PERFORMANCE EVALUATION

A. PERFORMANCE METRICS

This section outlines the criteria used to evaluate the effectiveness of the proposed strategy and presents the results from the conducted experiments. The true effectiveness of the detection techniques can only be validated through comprehensive testing and analysis. For this purpose, the experiments were carried out on a system with an Intel i7 processor, 16GB RAM, and a 2.31GHz CPU. The goal is to approximate the real-world accuracy of the methods employed. The evaluation involves two main types of detection errors: false positives (FP) and false negatives (FN). A false positive occurs when a detection system incorrectly classifies a non-event as an event, whereas false negatives are actual events that are incorrectly classified as non-events. Conversely, true negatives (TN) are correctly identified non-events, and true positives (TP) are accurately

Algorithm 2 Multistage Classification

Input: $X_{selected}$ – Preprocessed network traffic data with selected features
Output: Y_{attack} - Final attack classification
Steps:
// Stage 1: CapsNets for Initial Classification
1: Begin
2: Apply convolutional layers to $X_{selected}$
3: Generate feature maps F
4: for each feature map in F do
5: Process feature map through capsules
6: Produce capsule outputs U_{ij}
7: end for
8: for each capsule pair (i, j) do
9: Compute coupling coefficients c_{ij}
10: Compute capsule activations V_j
11: end for
12: Compute class probabilities Y from V_j
13: if Y indicates attack, then
14: $X_{attack} = X_{selected}$
15: else
16: Log traffic as normal
17: exit
18: end if
// Stage 2: Attention-Augmented RNN for Detailed Classification
19: Process X_{attack} through RNN
20: Obtain hidden states H_t for each time step t
21: for each time step t do
22: Compute attention scores α_t
23: end for
24: Compute context vector C_t using α_t
25: Classify attack using context vector C_t
26: Determine specific attack type Y_{attack}
27: end

detected events. Various metrics are used to assess the performance of the machine learning algorithms based on these classifications [45], [46].

Accuracy: This metric quantifies the ratio of correctly classified instances (including both true positives and true negatives) to the total number of instances. It is computed using the formula provided in (20).

$$Accuracy = \frac{TN + TP}{TN + TP + FN + FP} \tag{20}$$

Precision: Precision gauges the proportion of true positive classifications among all positive predictions made by the model. It is defined by the formula in (21).

$$Precision = \frac{TP}{TP + FP} \tag{21}$$

Recall: Recall measures the proportion of actual positive cases that the system correctly identified. It is computed using the formula provided in (22).

$$Recall = \frac{TP}{TP + FN} \tag{22}$$

F1 Score: The F1 score, which ranges from 0 to 1, represents the harmonic mean of precision and recall, providing a unified measure to evaluate their balance. It is computed as shown in (23).

$$F1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{23}$$

B. HYPERPARAMETER SETTING

The hyperparameters and training settings used in this experiment are detailed in Table 2.

TABLE 2. Hyperparameters.

Component	Hyperparameter	Optimal Value
QIPSO	Inertia Weight (ω)	0.7
	Cognitive Coefficient (c1)	1.8
	Social Coefficient (c2)	1.8
	Number of Iterations (T)	100-200
ANFIS	Number of Membership Functions	3-5
	Error Minimization Method	Gradient Descent
(CapsNets)	Routing Coefficients c_{ij}	0.1-0.9
Attention-Augmented RNN	Hidden State Size h_t	128-256

C. DATASET

The BoT-IoT dataset and the ToN-IoT dataset are both publicly available and widely used in network security research. These datasets are commonly used for IDS, particularly for IoT networks, as they contain a rich set of features and various types of attacks for model training and evaluation. The BoT-IoT dataset includes features such as timestamps, source and destination IP addresses, source and destination ports, protocols, flow duration, packet size, packet count, flow statistics, and labels indicating normal or attack traffic. Attack types in this dataset include DDoS, Botnet, and DoS. Similarly, the ToN-IoT dataset includes features like timestamps, IP addresses, ports, protocols, packet counts, byte counts, flow duration, flow statistics, and labels, with various attack types such as DDoS and Port Scan. Both datasets provide comprehensive traffic data that supports the training of anomaly detection models for IoT security, with labeled instances for supervised learning. These datasets are designed to capture both normal and malicious network behavior to help develop and evaluate intrusion detection systems.

BoT-IoT dataset: The BoT-IoT dataset encompasses both normal IoT network traffic and a variety of attack scenarios, making it suitable for evaluating our system. It was selected for its comprehensive representation of an IoT environment. The dataset includes attack types such as DoS, DDoS, data exfiltration, keylogging, service scanning, and OS-level attacks. Table 3 provides a summary of the number of benign and attack samples contained in the dataset.

TABLE 3. Statistics of attack and normal behavior in the BoT-IoT dataset.

Behavior Type	Number of Instances
Normal	477
OS fingerprinting	17914
Service scanning	73168
DoS TCP	615800
DoS HTTP	1485
DoS UDP	1032975
Data theft	6
Keylogging	73
DDoS UDP	948255
DDoS TCP	977380
DDoS HTTP	989

ToN-IoT dataset: The TON-IoT network dataset collects data from a diverse range of sources, including telemetry from IoT and Industrial IoT (IIoT) sensors, as well as network traffic and operating system data from both Windows and Ubuntu environments. The dataset contains a total of 22,339,021 instances, divided into two primary classes: one for differentiating between normal and attack data, and another featuring ten distinct categories. These categories consist of normal traffic and nine specific attack types: Backdoor, DDoS, DoS, Injection, Password, Ransomware, Scanning, XSS, and MITM. A total of 461,043 records are chosen for this analysis, incorporating both training and testing datasets. The selection of samples from the ToN-IoT dataset was based on both random sampling and the need for a balanced representation of normal and attack traffic. This selection ensures a thorough evaluation of the system's performance across normal conditions and multiple attack scenarios. Table 4 provides a summary of the selected data instances from the TON-IoT dataset.

TABLE 4. Summary of selected data instances from the ToN-IoT dataset.

Behavior Type	Number of Instances
Normal	300,000
Backdoor	20,000
DDoS	20,000
DoS	20,000
Injection	20,000
Password	20,000
Ransomware	20,000
Scanning	20,000
XSS	20,000
MITM	1043

D. DISCUSSION OF FINDINGS

The experiments conducted with varying data splits (60%-40%, 70%-30%, and 90%-10%) to evaluate the impact of SMOTE on model performance showed that the 80%-20% split, originally used, provided the best results. While the 60%-40% and 70%-30% splits offered improvements in class balance, the model experienced some issues with underfitting and overfitting. The 90%-10% split led to overfitting despite having more training data. Ultimately, the 80%-20% split struck the optimal balance, ensuring effective model training and evaluation on unseen data, which resulted in the best overall performance across key metrics like accuracy, precision, and recall.

The results clearly demonstrate that preprocessing techniques have a significant impact on the performance of the proposed model. Without any preprocessing, the model achieved an accuracy of 89.50%, precision of 85.00%, recall of 80.00%, F-measure of 82.50%, and AUC-ROC of 0.88. However, when standardization was applied, accuracy increased to 92.10%, precision to 88.00%, recall to 90.00%, F-measure to 89.00%, and AUC-ROC to 0.92. Normalization alone also resulted in improved performance, achieving an accuracy of 91.80%, precision of 87.50%, recall of 88.50%, F-measure of 88.90%, and AUC-ROC of 0.91. The inclusion of data cleaning further boosted the model, with accuracy reaching 93.00%, precision at 89.50%, recall at 92.00%, F-measure at 90.70%, and AUC-ROC at 0.93. Applying SMOTE enhanced performance even further, resulting in 94.00% accuracy, 91.50% precision, 94.00% recall, 92.60% F-measure, and AUC-ROC of 0.94. Combining standardization and normalization improved the results to 94.50% accuracy, 92.00% precision, 94.50% recall, 93.20% F-measure, and AUC-ROC of 0.95. The best performance was observed when using a combination of standardization, normalization, data cleaning, and SMOTE, achieving 98.20% accuracy, 96.00% precision, 98.00% recall, 97.00% F-measure, and AUC-ROC of 0.99.

After applying QIPSO + ANFIS for feature selection, the BOT-IoT dataset was reduced from 46 features to 22, while the TON-IoT dataset was reduced from 45 features to 20. The application of SMOTE addressed class imbalances, transforming the data distribution from highly skewed, with minority classes constituting as little as 2%, to an approximately equal distribution across all classes. In the BoT-IoT dataset, there are 36,68,522 original instances, with 73,400 synthetic samples generated using SMOTE, accounting for approximately 2% of the total data. The dataset is split into training and testing sets in an 80-20% ratio, resulting in 29,34,817 instances for training and 7,33,705 instances for testing. Of the synthetic data, 58,720 samples are in the training set, and 14,680 are in the testing set. The total combined dataset, including both original and synthetic data, consists of 3,741,922 instances, with 73,400 of them being synthetic. In the ToN-IoT dataset, there are 461,043 original instances, and 9,250 synthetic samples generated using SMOTE, making up approximately 2% of the total data. After splitting the dataset in an 80-20% ratio, the training data consists of 368,834 instances, and the testing data consists of 92,209 instances. The synthetic data is distributed with 7,400 samples in the training set and 1,850 in the testing set. The total combined dataset in the ToN-IoT dataset, including both original and synthetic data, has 470,293 instances, with 9,250 synthetic samples.

Table 5 summarizes the performance of the proposed system on the BoT-IoT dataset. The system recorded an overall accuracy of 98.96%, along with an average precision of

98.45%, an average recall of 98.63%, and an average F1-Score of 98.49%. This table illustrates the system's high efficacy in correctly identifying various attack types and benign traffic. Fig. 2 visually represents the performance metrics for various behavior types as shown in Table 4.

TABLE 5. System performance on attack types and benign traffic (BoT-IoT).

Behavior Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Benign	99.13	98.50	98.80	98.65
OS fingerprinting	98.41	98.10	98.22	98.10
Service scanning	98.35	97.90	98.12	98.13
DoS TCP	99.10	98.60	98.91	98.76
DoS HTTP	99.13	98.61	98.83	98.65
DoS UDP	98.91	98.54	98.70	98.55
Data theft	99.14	98.40	99.00	98.80
Keylogging	98.75	98.23	98.30	98.27
DDoS UDP	99.08	98.50	98.30	98.62
DDoS TCP	99.21	98.70	99.18	98.28
DDoS HTTP	99.32	98.81	99.08	98.91

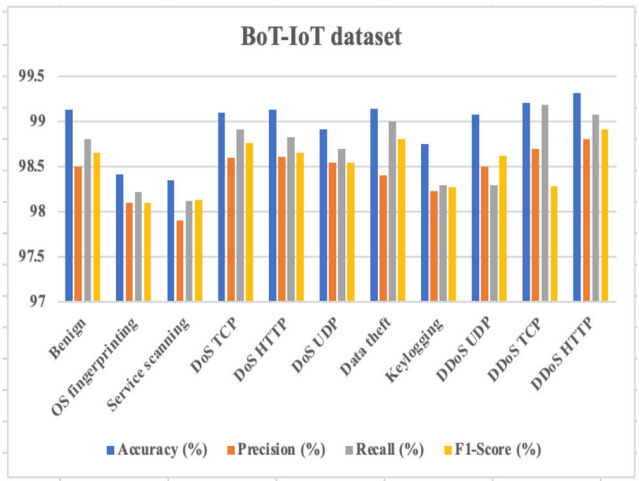

FIGURE 2. System performance on attack types and benign traffic (BoT-IoT dataset).

Table 6 displays the performance metrics of the proposed system evaluated on the TON-IoT dataset. The system attained an overall accuracy of 99.20% for normal traffic, with specific accuracy rates ranging from 98.45% for injection attacks to 99.15% for DDoS attacks. Additionally, precision, recall, and F1-Score values remained consistently high across all types of behavior. The average precision reached up to 99.10% for normal traffic, while the average recall peaked at 99.30% for normal traffic as well. The average F1-Scores varied but remained strong, reflecting a well-balanced performance in identifying both normal and malicious activities. Overall, the system demonstrated exceptional capability in accurately detecting different attack types and benign traffic, showcasing its effectiveness and reliability.

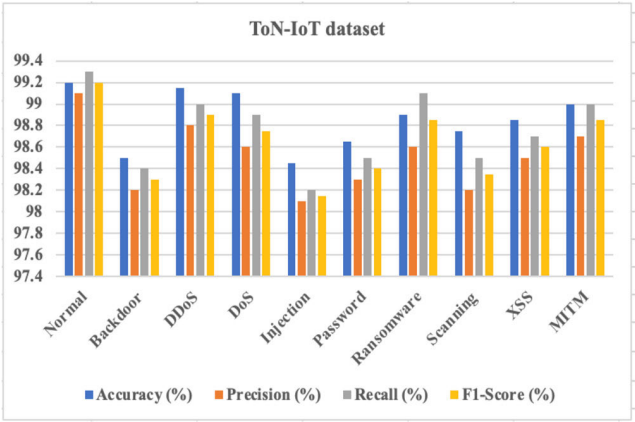

FIGURE 3. System performance on attack types and benign traffic (ToN-IoT dataset).

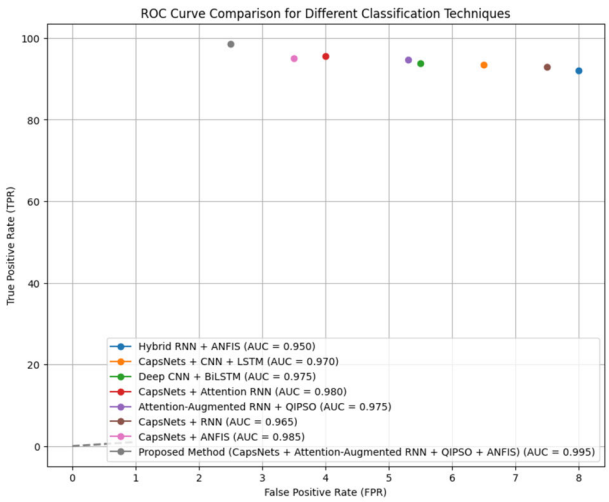

FIGURE 4. AUC-ROC of classification methods on the BoT-IoT dataset.

TABLE 6. System performance on attack types and benign traffic (TON-IoT).

Behavior Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Normal	99.20	99.10	99.30	99.20
Backdoor	98.50	98.20	98.40	98.30
DDoS	99.15	98.80	99.00	98.90
DoS	99.10	98.60	98.90	98.75
Injection	98.45	98.10	98.20	98.15
Password	98.65	98.30	98.50	98.40
Ransomware	98.90	98.60	99.10	98.85
Scanning	98.75	98.20	98.50	98.35
XSS	98.85	98.50	98.70	98.60
MITM	99.00	98.70	99.00	98.85

Fig. 3 illustrates the performance of the proposed system relative to different attack types and benign traffic on the TON-IoT dataset. It visually presents the system's accuracy, precision, recall, and F1-Score across various behavior types, highlighting its effectiveness and reliability in detecting both normal and malicious activities.

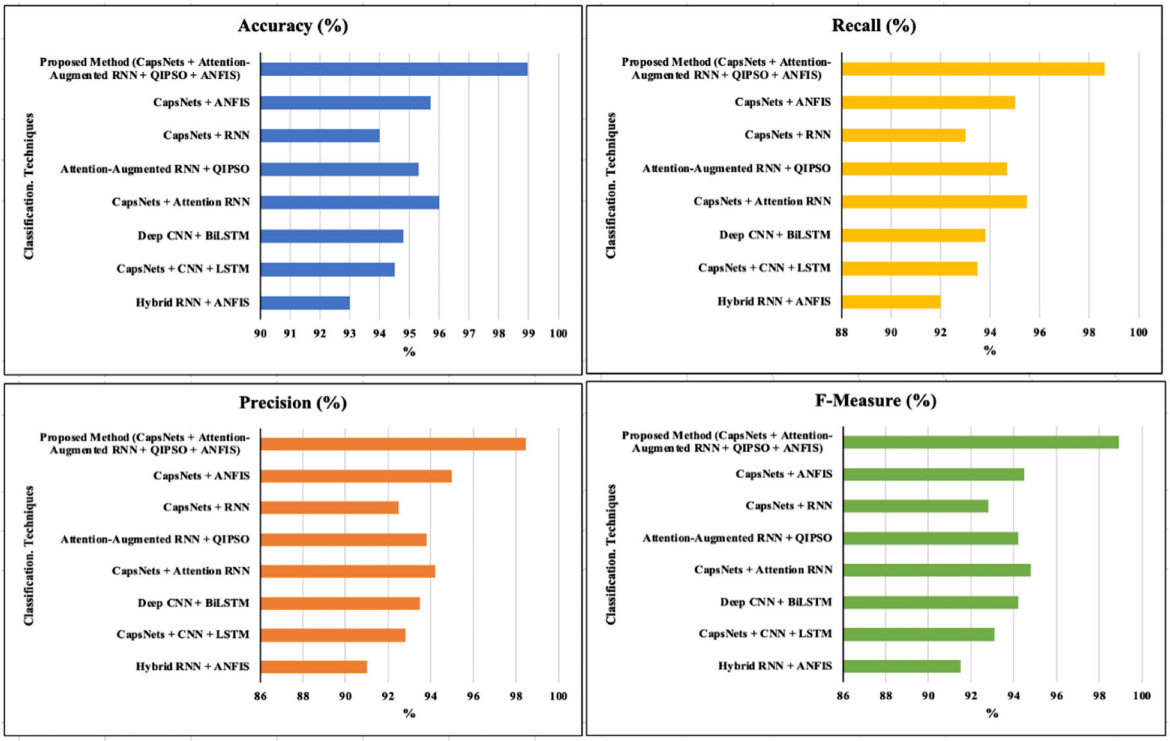

FIGURE 5. Evaluation of classification techniques on the BoT-IoT dataset.

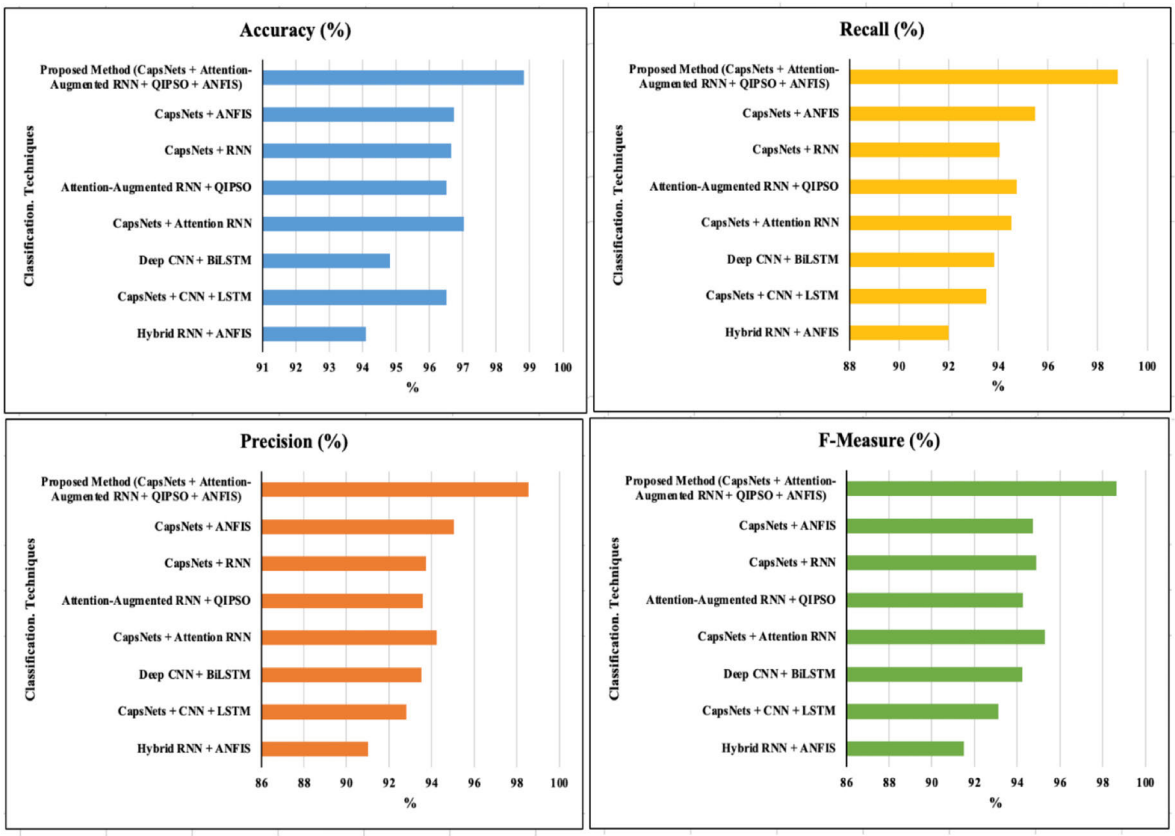

FIGURE 6. Evaluation of classification methods on the ToN-IoT dataset.

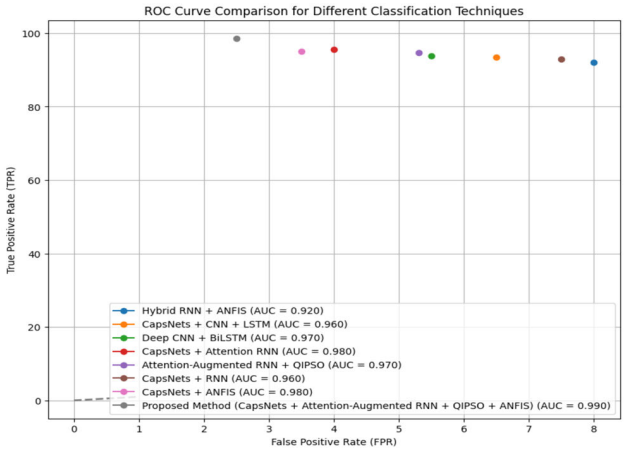

FIGURE 7. AUC-ROC of classification methods on the ToN-IoT.

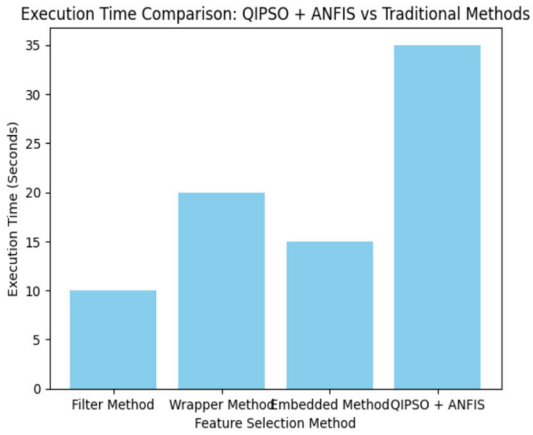

FIGURE 8. Execution time comparison.

TABLE 7. Evaluation of classification methods on the BoT-IoT dataset.

Classification Technique	Accuracy (%)	Precision (%)	Recall (%)	F-Measure (%)
Hybrid RNN + ANFIS [43]	93.0	91.0	92.0	91.5
CapsNets + CNN + LSTM	94.5	92.8	93.5	93.1
Deep CNN + BiLSTM [44]	94.8	93.5	93.8	94.2
CapsNets + Attention RNN	96.0	94.2	95.5	94.8
Attention-Augmented RNN + QIPSO	95.3	93.8	94.7	94.2
CapsNets + RNN	94.0	92.5	93.0	92.8
CapsNets + ANFIS	95.7	95.0	95.0	94.5
Proposed Method (CapsNets + Attention-Augmented RNN + QIPSO + ANFIS)	**98.6**	**98.5**	**98.63**	**98.94**

Table 7 provides a comparison of the performance of different classification techniques utilized for intrusion detection on the BoT-IoT dataset. For comparison purposes, the techniques found in related research were re-implemented to produce the results presented in Tables 6 and 7. The proposed method, which combines CapsNets, Attention-Augmented RNN, QIPSO, and ANFIS, outperforms all other approaches in every metric, achieving an accuracy of 98.96%, precision of 98.45%, recall of 98.63%, and an F-Measure of 98.94%. This highlights its effectiveness in accurately classifying and detecting network traffic anomalies compared to other techniques such as Hybrid RNN + ANFIS, CapsNets + CNN + LSTM, and Deep CNN + BiLSTM, which also show strong performance but fall short of the proposed method's metrics. Fig. 4 visually represents the comparative performance of classification techniques on the BoT-IoT dataset.

TABLE 8. Evaluation of classification methods on the ToN-IoT dataset.

Classification Technique	Accuracy (%)	Precision (%)	Recall (%)	F-Measure (%)
Hybrid RNN + ANFIS [43]	94.10	91.0	92.0	91.51
CapsNets + CNN + LSTM	96.51	92.8	93.51	93.12
Deep CNN + BiLSTM [44]	94.82	93.52	93.83	94.25
CapsNets + Attention RNN	97.03	94.24	94.52	95.28
Attention-Augmented RNN + QIPSO	96.52	93.58	94.74	94.26
CapsNets + RNN	96.65	93.73	94.06	94.88
CapsNets + ANFIS	96.74	95.06	95.50	94.75
Proposed Method (CapsNets + Attention-Augmented RNN + QIPSO + ANFIS)	**98.83**	**98.56**	**98.81**	**98.65**

The AUC-ROC curve in Fig. 6 offers a visual representation of the effectiveness of each classification technique in distinguishing between normal and attack behaviors. The proposed method shows the highest AUC-ROC and TPR indicating superior overall performance, as it maximizes the true positive rate while minimizing false positives and false alarms.

Figure 7 visually represents the superior performance of the Proposed Method (CapsNets + Attention-Augmented RNN + QIPSO + ANFIS), which achieves the highest AUC-ROC score of 0.99, demonstrating its effectiveness in classifying network traffic.

Table 8 presents a comparison of different classification techniques for intrusion detection on the ToN-IoT dataset. The proposed method, which integrates CapsNets, Attention-Augmented RNN, QIPSO, and ANFIS, achieves superior results with an accuracy of 98.83%, precision of 98.56%, recall of 98.81%, and an F-Measure of 98.65%. This highlights its outstanding effectiveness in accurately detecting and classifying network anomalies. In contrast, methods such as CapsNets + ANFIS and CapsNets + Attention RNN also demonstrate strong performance but fall short of matching the accuracy and balance across all metrics achieved by the

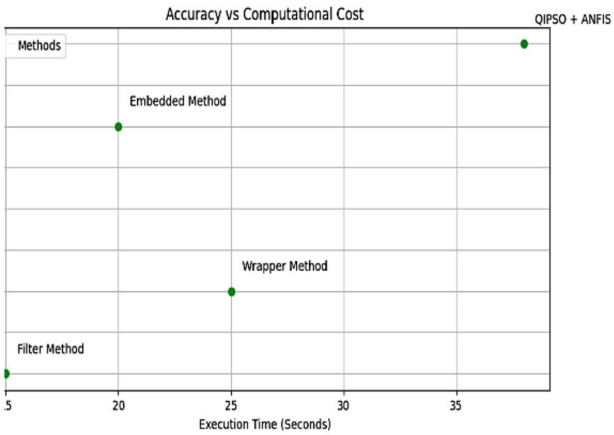

FIGURE 9. Accuracy vs computational Cos.

proposed approach. Fig. 5 provides a visual comparison of the performance of these classification techniques on the ToN-IoT dataset.

We also conducted experiments The QIPSO + ANFIS hybrid feature selection model demonstrated superior performance over the CapsNets + Attention-Augmented RNN approach in IoT anomaly detection using the ToN-IoT dataset. The QIPSO + ANFIS method achieved higher accuracy (98.3%), precision (96.8%), recall (97.5%), F-measure (97.15%), and AUC-ROC (0.99), significantly outperforming CapsNets + Attention-Augmented RNN, which had accuracy (96.1%), precision (94.5%), recall (94.0%), F-measure (94.2%), and AUC-ROC (0.97). The hybrid model's ability to efficiently select and refine relevant features, thanks to QIPSO's optimization and ANFIS's fuzzy logic, contributed to its enhanced detection capabilities, highlighting its advantages in handling complex IoT datasets.

The computational overhead of using QIPSO and ANFIS is higher compared to traditional feature selection methods such as filter, wrapper, and embedded approaches. This is primarily due to the iterative nature of QIPSO, which optimizes the feature subset, and ANFIS, which refines the relevance scores of selected features. As shown in Fig. 8, the resource consumption of QIPSO and ANFIS is greater due to the complex processes involved. However, as illustrated in Fig. 9. the hybrid approach of QIPSO and ANFIS significantly improves accuracy, effectively identifying more relevant features that enhance the overall performance of the IDS, particularly in complex IoT environments. While traditional methods are faster and less computationally demanding, they offer lower accuracy. Therefore, QIPSO and ANFIS provide a more effective solution for systems where high accuracy is critical, justifying the additional computational cost.

The training instability of Capsule Networks (CapsNets), attributed to the complex dynamic routing mechanism, was addressed by implementing careful weight initialization and gradient clipping. Furthermore, the computational complexity associated with both CapsNets and Attention-Augmented RNNs, which resulted in high memory consumption and prolonged training times, was alleviated through the adoption of efficient training strategies, including batch normalization, model size reduction, and pruning techniques. To mitigate the risk of overfitting, regularization methods such as dropout, early stopping, and data augmentation were incorporated. These measures ensured that the models not only performed effectively but also exhibited strong generalization capabilities on unseen data.

VI. CONCLUSION AND FUTURE WORK

In the face of increasing cyber threats targeting IoT networks, effective intrusion detection has become critical for maintaining the integrity and security of these systems. Conventional approaches frequently struggle to manage the complexity and dynamic characteristics of IoT ecosystems. The proposed IoT-based IDS presents a robust solution to enhance security in such environments by incorporating sophisticated preprocessing, hybrid feature selection, and multi-stage classification methods. By employing QIPSO for feature selection and refining it through the ANFIS, the system effectively filters out irrelevant features and improves overall detection accuracy. The multi-stage classification pipeline, incorporating CapsNets and Attention-Enhanced RNNs, allows the system to capture both layered data structures and time-based dependencies. This powerful framework excels at detecting a wide range of attack types, including rare and intricate anomalies. By utilizing this holistic approach, the system successfully tackles the distinct security challenges posed by IoT environments. However, despite the promising performance, there are some limitations that should be acknowledged. The limitation of the proposed IDS is its vulnerability to adversarial attacks, which can manipulate feature selection and classification, leading to misclassification and reduced detection performance. While adversarial training can improve robustness by introducing adversarial samples during training, it adds computational complexity and overhead. This increases training time and resource demands, which could be problematic for real-time or resource-constrained IoT environments, affecting the system's efficiency and scalability despite enhancing its resilience to attacks.

Future studies could aim at enhancing the system for real-time intrusion detection, with a specific focus on deployment in resource-limited IoT devices. Enhancements such as model compression, edge computing, and hardware acceleration could be explored to reduce computational overhead and improve response times. Additionally, scaling the system for larger IoT networks may require distributed architectures or federated learning approaches to manage increased data volumes and device diversity effectively. Continued exploration of strategies to defend against adversarial attacks will be essential for maintaining the robustness of the IDS. Techniques like adversarial training, anomaly detection, and the integration of more resilient model architectures should be explored. Finally, improving the system's generalization

capabilities to adapt to different IoT environments without extensive retraining could be achieved through transfer learning, enabling broader applicability in real-world scenarios. By addressing these future research directions, the proposed system could become a more powerful and adaptable solution for securing IoT ecosystems against evolving cyber threats.

REFERENCES

[1] W. Li, S. Tug, W. Meng, and Y. Wang, "Designing collaborative blockchained signature-based intrusion detection in IoT environments," *Future Gener. Comput. Syst.*, vol. 96, pp. 481–489, Jul. 2019.

[2] M. F. Elrawy, A. I. Awad, and H. F. A. Hamed, "Intrusion detection systems for IoT-based smart environments: A survey," *J. Cloud Comput.*, vol. 7, no. 1, pp. 1–20, Dec. 2018.

[3] B. B. Zarpelão, R. S. Miani, C. T. Kawakani, and S. C. D. Alvarenga, "A survey of intrusion detection in Internet of Things," *J. Netw. Comput. Appl.*, vol. 84, pp. 25–37, Feb. 2017.

[4] V. Kumar, A. K. Das, and D. Sinha, "UIDS: A unified intrusion detection system for IoT environment," *Evol. Intell.*, vol. 14, no. 1, pp. 47–59, Mar. 2021.

[5] J. C. Sicato, S. K. Singh, S. Rathore, and J. H. Park, "A comprehensive analysis of intrusion detection system for IoT environment," *J. Inf. Process. Syst.*, vol. 16, no. 4, pp. 975–990, 2020.

[6] K. Vaigandla, N. Azmi, and R. Karne, "Investigation on intrusion detection systems (IDSs) in IoT," *Int. J. Emerg. Trends Eng. Res.*, vol. 10, no. 3, pp. 158–166, Mar. 2022.

[7] C. Hazman, S. Benkirane, A. Guezzaz, M. Azrour, and M. Abdedaime, "Intrusion detection framework for IoT-based smart environments security," in *Proc. Int. Conf. Artif. Intell. Smart Environ.* Cham, Switzerland: Springer, Nov. 2022, pp. 546–552.

[8] C. Hazman, A. Guezzaz, S. Benkirane, and M. Azrour, "Toward an intrusion detection model for IoT-based smart environments," *Multimedia Tools Appl.*, vol. 83, no. 22, pp. 62159–62180, Aug. 2023.

[9] G. Luo, Z. Chen, and B. O. Mohammed, "A systematic literature review of intrusion detection systems in the cloud-basedIoTenvironments," *Concurrency Comput., Pract. Exper.*, vol. 34, no. 10, p. 6822, May 2022.

[10] J. Arshad, M. A. Azad, R. Amad, K. Salah, M. Alazab, and R. Iqbal, "A review of performance, energy and privacy of intrusion detection systems for IoT," *Electronics*, vol. 9, no. 4, p. 629, Apr. 2020.

[11] X. Fu, N. Zhou, L. Jiao, H. Li, and J. Zhang, "The robust deep learning–based schemes for intrusion detection in Internet of Things environments," *Ann. Telecommun.*, vol. 76, nos. 5–6, pp. 273–285, Jun. 2021.

[12] H. Safwan, Z. Iqbal, R. Amin, M. A. Khan, M. Alhaisoni, A. Alqahtani, and B. Chang, "An IoT environment-based framework for intelligent intrusion detection," *CMC Comput. Mater. Contin.*, vol. 75, pp. 2365–2381, Jan. 2023.

[13] N. Chaabouni, M. Mosbah, A. Zemmari, C. Sauvignac, and P. Faruki, "Network intrusion detection for IoT security based on learning techniques," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2671–2701, 3rd Quart., 2019.

[14] D. S. Smys, D. A. Basar, and D. H. Wang, "Hybrid intrusion detection system for Internet of Things (IoT)," *J. ISMAC*, vol. 2, no. 4, pp. 190–199, Sep. 2020.

[15] M. S. A. Muthanna, R. Alkanhel, A. Muthanna, A. Rafiq, and W. A. M. Abdullah, "Towards SDN-enabled, intelligent intrusion detection system for Internet of Things (IoT)," *IEEE Access*, vol. 10, pp. 22756–22768, 2022.

[16] S. Hajiheidari, K. Wakil, M. Badri, and N. J. Navimipour, "Intrusion detection systems in the Internet of Things: A comprehensive investigation," *Comput. Netw.*, vol. 160, pp. 165–191, Sep. 2019.

[17] N. Islam, F. Farhin, I. Sultana, M. S. Kaiser, M. S. Rahman, M. Mahmud, A. S. M. S. Hosen, and G. H. Cho, "Towards machine learning based intrusion detection in IoT networks," *Comput., Mater. Continua*, vol. 69, no. 2, pp. 1801–1821, 2021.

[18] C. Liang, B. Shanmugam, S. Azam, M. Jonkman, F. D. Boer, and G. Narayansamy, "Intrusion detection system for Internet of Things based on a machine learning approach," in *Proc. Int. Conf. Vis. Towards Emerg. Trends Commun. Netw. (ViTECoN)*, Mar. 2019, pp. 1–6.

[19] A. Nazir, Z. Memon, T. Sadiq, H. Rahman, and I. U. Khan, "A novel feature-selection algorithm in IoT networks for intrusion detection," *Sensors*, vol. 23, no. 19, p. 8153, Sep. 2023.

[20] V. P. Sharma, N. S. Yadav, S. S. Adavi, D. S. D. Reddy, and B. B. Gupta, "A two stage hybrid intrusion detection using genetic algorithm in IoT networks," *J. Discrete Math. Sci. Cryptogr.*, vol. 26, no. 3, pp. 667–676, 2023.

[21] P. Sanju, "Enhancing intrusion detection in IoT systems: A hybrid metaheuristics-deep learning approach with ensemble of recurrent neural networks," *J. Eng. Res.*, vol. 11, no. 4, pp. 356–361, Dec. 2023.

[22] M. Nivaashini, P. Thangaraj, S. Sountharrajan, E. Suganya, and R. S. Soundariya, "Effective feature selection for hybrid wireless IoT network intrusion detection systems using machine learning techniques," *Ad Hoc Sens. Wireless Netw.*, vol. 49, pp. 175–206, Jan. 2021.

[23] S. Walling and S. Lodh, "Network intrusion detection system for IoT security using machine learning and statistical based hybrid feature selection," *Secur. Privacy*, vol. 7, no. 6, p. e429, Nov. 2024.

[24] V. Hajisalem and S. Babaie, "A hybrid intrusion detection system based on ABC-AFS algorithm for misuse and anomaly detection," *Comput. Netw.*, vol. 136, pp. 37–50, May 2018.

[25] M. A. Khan, N. Ahmed, J. Padela, M. S. Raza, A. Gangopadhyay, J. Wang, J. Foulds, C. Busart, and R. F. Erbacher, "Flood-ResNet50: Optimized deep learning model for efficient flood detection on edge device," in *Proc. Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2023, pp. 512–519.

[26] M. A. Shyaa, Z. Zainol, R. Abdullah, M. Anbar, L. Alzubaidi, and J. Santamaría, "Enhanced intrusion detection with data stream classification and concept drift guided by the incremental learning genetic programming combiner," *Sensors*, vol. 23, no. 7, p. 3736, Apr. 2023.

[27] K. Wu, Z. Chen, and W. Li, "A novel intrusion detection model for a massive network using convolutional neural networks," *IEEE Access*, vol. 6, pp. 50850–50859, 2018.

[28] A. G. Ayad, N. A. Sakr, and N. A. Hikal, "A hybrid approach for efficient feature selection in anomaly intrusion detection for IoT networks," *J. Supercomput.*, vol. 80, no. 19, pp. 26942–26984, Dec. 2024.

[29] A. Guerra-Manzanares, H. Bahsi, and S. Nomm, "Hybrid feature selection models for machine learning based botnet detection in IoT networks," in *Proc. Int. Conf. Cyberworlds (CW)*, Oct. 2019, pp. 324–327.

[30] Z. Liu and Y. Shi, "A hybrid IDS using GA-based feature selection method and random forest," *Int. J. Mach. Learn. Comput.*, vol. 12, no. 2, pp. 43–50, 2022.

[31] E. M. Maseno and Z. Wang, "Hybrid wrapper feature selection method based on genetic algorithm and extreme learning machine for intrusion detection," *J. Big Data*, vol. 11, no. 1, pp. 1–25, Feb. 2024.

[32] E. O. O. Okpu, O. E. Taylor, N. D. Nwiabu, and D. Matthias, "A hybrid machine learning approach for intrusion detection and mitigation on IoT smart healthcare," *Int. J. Adv. Comput. Sci. Technol.*, vol. 13, no. 7, pp. 82–90, 2024.

[33] S. Gurumurthy, D. Pamela, and U. Roy, "Hybrid pigeon inspired optimizer-gray wolf optimization for network intrusion detection," *J. Syst. Manage. Sci.*, vol. 12, no. 4, pp. 383–397, 2022.

[34] J. S. Prasath, V. I. Shyja, P. Chandrakanth, B. K. Kumar, and A. Raja Basha, "An optimal secure defense mechanism for DDoS attack in IoT network using feature optimization and intrusion detection system," *J. Intell. Fuzzy Syst.*, vol. 46, no. 3, pp. 6517–6534, Mar. 2024.

[35] M. Sajid, K. R. Malik, A. Almogren, T. S. Malik, A. H. Khan, J. Tanveer, and A. U. Rehman, "Enhancing intrusion detection: A hybrid machine and deep learning approach," *J. Cloud Comput.*, vol. 13, no. 1, p. 123, Jul. 2024.

[36] R. K. Agrawal, B. Kaur, and P. Agarwal, "Quantum inspired particle swarm optimization with guided exploration for function optimization," *Appl. Soft Comput.*, vol. 102, Apr. 2021, Art. no. 107122.

[37] L. Wang, M. Zhang, Q. Niu, and J. Yao, "A modified quantum-inspired particle swarm optimization algorithm," in *Proc. 3rd Int. Conf.*, Taiyuan, China. Berlin, Germany: Springer, Sep. 2011, pp. 412–419.

[38] M. N. M. Salleh, N. Talpur, and K. Hussain, "Adaptive neuro-fuzzy inference system: Overview, strengths, limitations, and solutions," in *Data Mining and Big Data: Second International Conference, DMBD 2017, Fukuoka, Japan, July 27–August 1, 2017, Proceedings 2*. Cham, Switzerland: Springer, 2017, pp. 527–535.

[39] N. Ahmad, M. Hanmandlu, and M. F. Azeem, "Generalization of adaptive neuro-fuzzy inference systems," *IEEE Trans. Neural Netw.*, vol. 11, no. 6, pp. 1332–1346, Nov. 2000.

[40] O. El Alaoui-Elfels and T. Gadi, "From auto-encoders to capsule networks: A survey," in *Proc. E3S Web Conf.*, vol. 229, 2021, p. 1003.

[41] J. Gu, V. Tresp, and H. Hu, "Capsule network is not more robust than convolutional network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14304–14312.

[42] J. Tan and J. Oyekan, "Attention augmented convolutional neural network for acoustics based machine state estimation," *Appl. Soft Comput.*, vol. 110, Oct. 2021, Art. no. 107630.
[43] J. Liu, W. Yinchai, T. C. Siong, X. Li, L. Zhao, and F. Wei, "On the combination of adaptive neuro-fuzzy inference system and deep residual network for improving detection rates on intrusion detection," *PLoS ONE*, vol. 17, no. 12, Dec. 2022, Art. no. e0278819.
[44] S. Sadhwani, M. A. H. Khan, R. Muthalagu, and P. M. Pawar, "BiLSTM-CNN hybrid intrusion detection system for IoT application," Jun. 2024, doi: 10.21203/rs.3.rs-3820775/v1.
[45] G. Barseghyan, Y. Yuan, and M. Anakpa, "Model for detection of masquerade attacks based on variable-length sequences," *IEEE Access*, vol. 8, pp. 210140–210157, 2020.
[46] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

G. LOGESWARI received the B.E. degree in computer science and engineering and the M.Tech. and Ph.D. degrees in information technology from Anna University, Chennai, India, in 2012, 2014, and 2024, respectively. She is currently an Assistant Professor (Senior) with the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai. With nine years of professional experience in teaching and research, she has published over 30 papers in esteemed international journals and conferences. Her primary research interests include software-defined networks, the Internet of Things, machine learning, deep learning, and information security.

J. DEEPIKA ROSELIND receive the Ph.D. degree in computer vision, in 2023. She is an Assistant Professor (Senior Grade) with Vellore Institute of Technology, Chennai Campus. She has over ten years of experience in teaching. Her major field of study includes machine learning, the Internet of Things, and human–computer interaction.

K. TAMILARASI received the Master of Engineering and Ph.D. degrees in computer science from Anna University, Chennai, in 2010 and 2020, respectively. She holds a distinguished academic career. She is currently an Assistant Professor (Senior Grade-2) with the School of Science and Engineering, VIT University, Chennai Campus, with more than 15 years of extensive experience in academia. Her areas of expertise encompass cloud computing, artificial intelligence, database management systems, and healthcare analytics. She has made significant contributions to her fields of specialization, with over 20 research papers published in reputed international journals and conference proceedings.

V. NIVETHITHA received the M.Tech. degree from Puducherry Technological University and the Ph.D. from the National Institute of Technology Puducherry (NITPY), specializing in edge computing, specifically focusing on data placement, task scheduling, and optimization in network environments. She is an Assistant Professor with VIT, with over a year of experience in academia. Her research interests include optimization and combinatorial problems within network systems, where she has contributed to advancing methodologies in data scheduling and placement.